

Reimagining Informed Consent in Human-Robot Interaction: Introducing the RoboConsent Framework

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Abstract—Informed consent is an integral process in human-robot interaction (HRI); however, current practices have been criticized for overlooking the social, psychological, and embodied complexities of interacting with robots. Social robots’ embodied, human-like design and social behavior can lead to misaligned expectations that pose risks such as deception, overtrust, poor user experience, and psychological harm for users. Moreover, robots often collect personal data in ways that are not always visible or understood by users. Typically, informed consent does not address such issues, highlighting the need for consent processes tailored to HRI. In this paper, we reimagine informed consent and introduce the RoboConsent framework, drawing from previous HRI research highlighting these issues and feminist consent models that address power imbalances and move toward a user-centered process. The framework consists of five components that ensure meaningful informed consent and six principles that guide how it can be obtained. These work in tandem to create informed consent practices that address the unique dynamics of HRI.

I. INTRODUCTION

Most people interact with technology daily, in both their personal and professional lives. These tools are sometimes simple to use and understand, allowing for informed decisions about when and how to use them. More often than not, however, such tools and devices have capabilities that individuals are not aware of, and thus they agree to partake in their use without all information needed. This is especially true with robots, which can sometimes be designed to look and behave in ways that mask their true abilities.

Informed consent is a process where individuals agree to participate in a situation, freely and without coercion, after receiving all relevant information needed to make an informed decision. It is a fundamental principle in ethics and law to safeguard individuals’ autonomy and personal rights in a given situation [1], [2]. There are multiple regulations and laws that deal with ethical issues and technology use, such as data and privacy concerns in EU’s General Data Protection Regulation (GDPR) [3] and Canada’s Personal Information Protection and Electronic Documents Act (PIPEDA) [4]; or ethical concerns related to Artificial Intelligence (AI) systems in the EU’s AI Act [5]. While Human-Robot Interaction (HRI) overlaps with the use of AI, especially with the growing integration of LLM “scarecrows” in robots [6], there are many aspects of consent in HRI that are not fully addressed in these regulations, such as human-like design, physical embodiment, and autonomy [1], [7], [2].

Individuals interacting with robots often lack the information needed to give meaningful informed consent, partly because of misaligned expectations due to the lack of direct first-hand experience with robots [8]. Users may expect robots to “work” like humans, concealing crucial aspects needed for informed consent. For example, anthropomorphic traits in robots (e.g., appearance and behavior) can conceal that they are collecting, storing, and sharing data, while encouraging users to disclose sensitive personal information [1], [2]. This can lead to a range of negative outcomes, including overtrust, particularly when human-like design causes users to rely on robots in ways that exceed their true abilities [9], [1]. Furthermore, manipulative design practices may exploit user vulnerabilities, resulting in psychological harm, emotional bonds, or feelings of betrayal when robots fail to meet expectations [10], [11].

Therefore, meaningful informed consent is a key goal in HRI that should be accomplished through ensuring that individuals fully understand all components of the interaction, including aligning their expectations with the robot’s actual capabilities [1]. To mitigate the risks of uninformed consent, it is essential to align user expectations with the robot’s capabilities, and include ethical safeguards throughout every stage of the interaction, ultimately improving the user experience and the ethical soundness of HRI [12]. Importantly, we argue that the responsibility of ensuring that consent is meaningful and informed should not fall solely on users, but rather, robots and their designers must play an active role in managing expectations and fostering mutual understanding [13].

This paper reimagines informed consent for HRI. We introduce RoboConsent, a framework of five components and six principles designed to establish ethically robust informed consent processes tailored to HRI. The framework is grounded in the idea that users are not a homogeneous group and is intended to support diverse needs. Drawing from feminist consent practices, we advocate for consent that is ongoing and freely given, ensuring it is more than a mandatory step but a continuous, user-centered process. This approach not only helps manage user expectations for meaningful informed consent but also enhances autonomy and trust while addressing underlying power imbalances [14], [15], [16], [17]. Our work builds on prior research on informed consent in HRI [2], [1], [7], [12] by integrating these.

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II. BACKGROUND

Informed consent can apply to a wide range of situations, with laws and regulations targeting different aspects of consent such as: agreeing to medical treatments or procedures [18], allowing the collection and use of personal data [4], understanding workplace surveillance [19], understanding the reasons for detention or arrest [20], agreeing to be photographed or recorded for public use [4], and participating in scientific research [21]. While these principles are similar in ensuring individuals are fully informed, they vary to address the risks, ethical concerns, and power dynamics that are at play in these different domains. Consenting to a medical procedure has potential consequences that are life threatening, whereas workplace surveillance has potential consequences related to the right to privacy in various forms.

Historically, informed consent meant making sure information was disclosed to the individual(s) involved, and over time this came to mean that the information disclosed should also be *understood* [22]. Moving beyond making sure an individual *understands* the information in order to consent, modern discourse on sexual consent emphasizes the addition of enthusiastic and ongoing consent [15]. These models touch on deeper ethical concerns on sexual violence and coercion, highlighting the significance of power dynamics and the need for individuals to have clear, voluntary, and informed consent over their participation [14]. Sexual consent models like FRIES, developed by the US advocacy group Planned Parenthood [23], advocate for consent to be Freely given, Reversible, Informed, Enthusiastic, and Specific.

Beyond the scope of sexual consent, feminist sexual consent models have been proposed for other domains as they address many of the issues raised relating to power dynamics and inequality, including user interaction and experience with technology [15], [16]. These models emphasize the importance of consent as an ongoing, enthusiastic, and negotiated process rather than a one-time agreement. Although current feminist models have been critiqued as being overly simplistic, overly binary, reinforcing gender stereotypes, and heteronormative [16], they remain an important step towards addressing power dynamics and ensuring more inclusive, ethical frameworks for consent. Their emphasis on agency and negotiated interactions makes them particularly relevant to discussions of informed consent in technology.

In Human-Computer Interaction (HCI), for example, issues related to consent include (1) the constant user data collection that occurs without users' knowledge, (2) persuasive design that aims to influence users' behavior and their attitudes, and (3) overwhelming users with information during the consent process which leads to "consent fatigue" [15], [24]. When consent is obtained in HCI, it is typically one single occurrence by accepting a lengthy licensing agreement, which is "by design one-sided, non-negotiated, and non-negotiable" [15, p.4]. These concerns extend beyond HCI and exist in adjacent technology fields, such as Artificial Intelligence (AI) [16]. While there are attempts to address consent issues in AI, like EU's AI Act

[5], it falls short of addressing broader issues that exist within AI today [16]. The AI act [5] sets clear boundaries on what is permitted during the testing of AI systems, but primarily frames consent as a one-time transaction, without promoting the need for an ongoing, contextual, and community driven consent process. Authors like Klein and D'Ignazio [16] argue for a feminist consent framework that can mitigate harmful power dynamics that AI can foster and integrate concepts like queer, collective, and interdependent consent.

These limitations in technology fields like HCI and AI highlight the need to rethink how consent is approached in other technology use as well. As robots are envisioned to enter personal and social spaces, these issues are amplified, requiring a reimagined approach to informed consent in HRI.

A. Informed consent in HRI: practical and legal challenges

The use of robots intersects with multiple ethical legal frameworks, including those addressing data and privacy (e.g., robots collecting personal data from users [4], [3]) and workplace regulations (e.g., healthcare professionals sharing workspaces with robots [19]). However, these frameworks lack crucial aspects relevant to HRI.

Robots intended for interactions with users are designed with the objective of acting, to some degree, autonomously within their environment [7], raising unique ethical issues related to privacy and loss of control [1]. As HRI works towards personalized robots, the demand to collect personal data - which may not be collected in a way that is obvious to users - increases. It may be hard for individuals to distinguish between personal data and non-personal data, such as a robot vacuum scanning their homes to map it, but ultimately capturing personal details about the users [7]. Robots that are designed for public spaces may collect and process data from multiple individuals simultaneously, often without explicit consent, which complicates the informed consent process further [25]. Moreover, because robots have a physical presence and can act autonomously, users may struggle to understand and expect their movements, leading to a sense of lost control during the interaction [1]. Ultimately, these are ethical issues, where individuals' own autonomy and self-ownership are compromised [1]. These challenges highlight the need to carefully consider how informed consent is obtained when users engage with autonomous robots across various contexts, encompassing both physical intervention and personal data processing by robots [25].

Building on these concerns, Sarathy et al. [12] argues that consent should be a distinct and critical area in HRI. Research areas like touch, proxemics and sex robots bring questions of consent to the forefront as they ultimately deal with social norms and what users find acceptable. For example, should a robot waiter be allowed to infer consent from implicit cues before clearing a table? The authors [12] suggest looking at consent from three different perspectives: actual (i.e., explicitly agrees to an action), apparent (i.e., user behavior suggests agreement), and presumed consent (i.e., consent is inferred in the absence of explicit refusal). Although presumed consent may violate social norms, like

pushing the user, it may be considered valid if it is in situations where the robot is removing someone out of harm's way. Thus, this suggests that robots should be designed to be able to interpret and respond to consent dynamically, based on the social and physical context. This kind of feature should be considered at the earliest stages of the design to proactively ask how consent applies to specific situations. By doing so, we can ensure that robots that are operating in shared spaces with humans, align with humans' expectations and ethical considerations.

Lintvedt [7] proposes models for administering consent, particularly in sensitive contexts like healthcare. They suggest that healthcare robots should operate under a two-phase consent model, where the first step involves healthcare professionals obtaining informed consent for the use of the robot, and the second step involves the robot itself obtaining consent related to data and privacy. The latter should be dynamic and ongoing, ensuring consent is obtained for different stages of data processing. It must be ensured that users can ask additional questions and receive further information from the robot as needed to provide informed consent [7]. It is possible that users may be intimidated by a robot due to its physical design, perceived intelligence, or a general fear of AI [7], which could introduce a harmful power imbalance that undermines the voluntariness of consent.

Building on these ideas, Winkle et al. [17] argue that informed consent in HRI is deeply entangled with questions of power and user agency, where questions of whether consent has been freely given or if it's been subtly coerced require a feminist lens to address such power imbalances. For example, patients interacting with healthcare robots may feel pressured to comply with requests due to perceived authority, or workers under robotic surveillance may not feel they have the agency to refuse certain interactions. The authors argue that, rather than being a one-time agreement, informed consent should be an ongoing, negotiated process that challenges systemic inequalities and prioritizes user agency. Taking these power dynamics into account raises the question of how robots should actually approach consent in real-life situations where human oversight and intervention might be limited.

Together, these perspectives highlight that informed consent in HRI is not only about data privacy but also about respecting user autonomy and addressing social and ethical complexities, calling for careful and ongoing attention in how we design and deploy robots. To address this, we argue that the design of robots and the consent procedure must be transparent and free from manipulative tactics. Ensuring that consent is genuinely voluntary is crucial to maintaining ethical standards in human-robot interactions [7], [17]. The above research, which emphasizes informed consent, is also deeply shaped by how users interpret and understand robotic capabilities. The next section explores psychological concerns with informed consent. Specifically, we explore how expectations affect users' ability to give meaningful consent.

B. The impact of expectations on informed consent in HRI

In line with existing research, we argue that current regulations and ethical frameworks on informed consent largely fail to consider the ways in which users' expectations of robots impact their capacity to consent meaningfully. Even if informed consent is *legally* sound, crucial aspects may be overlooked if users' expectations are not addressed, and risk leading to deception, overtrust, and psychological harm for the users [9], [26], [2].

Users' expectations are shaped by multiple factors, including the robot's appearance and behavior, users' previous experiences, and the context in which the interaction occurs [8]. Previous experience with robots directly influences how users interpret robot behavior and capabilities [8]. Managing these expectations is crucial, as mutual understanding in HRI requires robots to adapt to the individual's needs, rather than the burden falling entirely on users [13]. For example, users who have interacted with voice assistants like Siri may expect difficulty being understood by a robot due to speaking with an accent that deviates from an American male voice [27], [28], or those primarily familiar with fictional portrayals of robots might experience heightened fear of robots and their expected capabilities [29], [13].

In particular, what makes robots stand out from many other technological artifacts is their embodied and human-like design, which often leads to anthropomorphic interpretations and expectations of certain behaviors [30]. This design choice is intentional and aimed at creating successful interactions, but it also raises ethical questions; for example, robots emulating human-like emotions can be argued to be deceptive and manipulative [26]. This kind of manipulation does not only occur in controlled research environments but also in in-the-wild context as well, where informed consent is much harder to control due to limited human oversight. Robots designed for companionship, customer service, or healthcare may be used in a similar way to influence individuals in everyday contexts [12]. If users are not aware that the robot collects and stores data shared during the interaction, or that their emotional responses are being manipulated, they cannot give fully informed consent [2]. From this perspective, informed consent is not just about agreeing to participate but about understanding what one is consenting to, including the robot's functionalities and the ways in which the interaction may affect them. Although deception is part of many ethical standards like the Declaration of Helsinki [31], current ethical frameworks often overlook how users' expectations of robots shape these dynamics. As a result, unique forms of deception and manipulation in HRI, and the resulting harm in HRI remain insufficiently addressed [26], [32].

Ultimately, misaligned expectations of a robot's perceived capabilities can foster overtrust, which is especially concerning in high-stakes contexts such as healthcare [9]. This misplaced trust may cause users to rely too heavily on robots, leaving them vulnerable to errors or deceptive practices. In such cases, overtrust can undermine safety and degrade

the trustworthiness of human-robot interactions, exacerbating the ethical concerns associated with human-like design [11]. Recognizing this, some researchers have emphasized the importance of transparency and expectation management as part of the consent process. Fronemann et al. [1] emphasize that ensuring users are fully informed and have consented to the interaction is crucial, not only for safeguarding autonomy and data privacy, but also for aligning user expectations with robots' behavior and capabilities. Transparent design, explainability, and clear communication of data practices can help users build aligned expectations of the robot's capabilities and limitations [1].

Explainability is often used in HRI as a way to make robots more understandable for users [33], however, explainability may not always mean aligned expectations. A study by Zhong et al. [34] revealed that incorporating explainability features does not necessarily improve users' understanding of how the robot makes decisions. In addition, explainability often provides the same type of information to all users, regardless of their individual needs; however, users' own unique background and experiences with robots will affect their expectations and in turn their interactions [8]. Consequently, to effectively help users comprehend a robot's actions, we argue for a continuous and ongoing informed consent process. This requires a thorough understanding of users' initial and evolving expectations, which could potentially increase the explainability of the robot as well.

Other attempts to address expectations are through participatory approaches that involve users in shaping how robots communicate and behave. For example, Gebelli et al. [33] proposed a participatory design framework for explainability and transparency in HRI. Their approach emphasizes the co-design of explainable systems, engaging users directly in the design process to ensure that their expectations and needs are integrated into the robot's behavior up front, rather than added after its deployment. By refining interactions over time and incorporating explainability as a core design principle, the co-design process aims to align user expectations with the robot's functionality, creating trust, understanding, and a more intuitive user experience [33]. From this perspective, we contend that informed consent could be an extension of explainability, helping to reduce manipulation and power imbalances while increasing trust and aligned expectations.

III. THE ROBOSSENT FRAMEWORK

In this section, we present the RoboConsent framework (fig. 1), a tool for structuring more meaningful informed consent in HRI. The framework consists of five components that outline *what* users should be informed about, and six principles that describe *how* consent should be obtained and maintained. In other words, the components define the requirements of consent, while the principles guide the process by which that consent is obtained and maintained. The framework is grounded in the argument that informed consent in HRI must go beyond legal formalities to meaningfully engage with users' expectations, experiences, and concerns. When users' expectations about a robot's

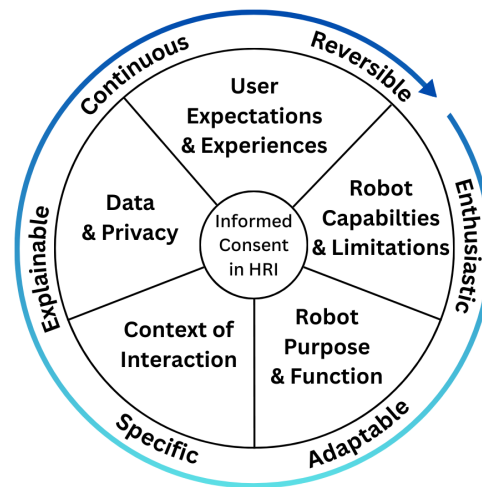


Fig. 1. The RoboConsent framework for meaningful informed consent in HRI. The circle consists of five components users must be informed about, while the surrounding labels consists of six guiding principles (CREASE) which describe how consent should be obtained and maintained throughout the interaction.

capabilities or intentions are misaligned with reality, consent may fail to provide the clarity or protection it is intended to offer and risk manipulation, overtrust, or psychological harm. RoboConsent aims to support more intentional, reflective, and responsive consent practices by drawing attention to both the substance and delivery of consent interactions in HRI.

A. Components of informed consent in HRI

We identify five core components that define requirements for informed consent in HRI, which cover what individuals should be aware of before, during, and after an interaction with a robot to be able to truly and knowingly consent to the interaction. These components were developed in response to concerns raised in prior work that informed consent is often treated as a one-dimensional event, rather than as a multifaceted process that reflects the complexity of HRI [12], [17], [25]. The five components presented here highlight key areas that the consent process must address, and that designers and roboticists need to consider. Although these components overlap in practice, we separate them here to draw attention to the distinct ethical and practical concerns each one raises. They build on prior work in HRI that emphasizes the importance of addressing user expectations and previous experience [12], [1], [8], robot capabilities and limitations [9], [12], robot purpose and function [7], context of interaction [25], [17], and data and privacy [2].

1) *Consider users' individual expectations and experiences*: Informed consent must take into account what users' expectations are and how they are shaped by their past experiences with technology and robots. These beliefs influence how people interpret and respond to a robot's behavior throughout the interaction. Thus, the informed consent process should recognize and manage users' previous experience and expectations with robots – before, during, and after the interaction. To obtain true informed consent,

the ongoing process should be tailored to each individual user, including how expectations are managed [8], [2]. This is also important for deciding what actions each individual will find acceptable in a given situation, especially as robots may need to navigate different types of consent, depending on the context and user [12]. Addressing people's individual experiences and expectations also aids in determining when explicit, or actual [12], consent is needed, and what types of actions might be acceptable within the interaction for each user. Incorporating this component in the informed consent process makes it possible to align the consent process with users' social norms and comfort levels, reducing the risk of overstepping boundaries during the interaction and avoiding feelings of loss of control [12], [1].

2) *Clearly communicate robot capabilities and limitations:* Informed consent must clearly communicate the robot's actual social and physical abilities, what it can and cannot do, so users understand what they are agreeing to, in order to support met expectations [8]. The multifunctional, autonomous, and sometimes deceptive nature of robots makes this component particularly pressing in HRI [26]. Unlike other types of technology, robots are able to act socially and physically, which makes it easier to fake some of their capabilities, or for people to believe that their capabilities are greater than they actually are [1], [12]. To avoid deception, whether intentional or otherwise, the robot's capabilities must therefore be known to the user. Capabilities can refer to cognitive, physical, and emotional abilities (or lack thereof), all of which should be addressed in informed consent. When users are not aware of the robot's actual capabilities, research suggests that they may overtrust robots, and rely on them in ways that are not appropriate or safe, especially in high-stakes contexts such as emergency scenarios or healthcare [9].

3) *Clearly indicate the robot's purpose and function:* Informed consent must make the robot's intended role and function transparent, including any secondary purposes that may not be immediately visible to users. Informed consent must clearly communicate why the robot is present, what it is supposed to do, and who might benefit from its use in that particular context. Although purpose and function are often addressed in traditional informed consent processes, they deserve particular attention in HRI because of how robots can serve multiple roles (e.g., social companionship and collecting personal data), some of which may not be immediately apparent to users [1]. Clarifying the robot's purpose and function is also essential to avoid misleading users about the nature of their interaction, and for addressing power imbalances that may arise from a robot's assigned role [17]. Making the robot's function explicit is not only about informing users of its role but also about protecting autonomy and preventing undue or unintentional influence during the interaction. It is also important to recognize that a robot's role or function may shift over time, even within the same context.

4) *Consider the context of the interaction:* The process and information provided for informed consent must be

shaped by the specific context in which the interaction takes place, as each setting influences user expectations, power dynamics, and other factors. What behaviors are considered acceptable for a robot to perform can vary depending on the context [25], [7]. It may be acceptable for a healthcare robot to collect health-related data in a hospital, but not in a private home. As Sarathy et al. [12] argue, consent must be understood in relation to the specific social and physical context, as what users perceive as acceptable or intrusive will vary depending on the situation. Therefore, context not only shapes the user's expectations of the robot but also defines the social norms that exist in that context. Consent processes should be designed to reflect these differences and adjust the level of disclosure, explanation, and user choice accordingly.

5) *Communicate data and privacy concerns:* Informed consent must explicitly address how the robot collects, processes, stores, and shares user data, especially when these practices are not immediately visible. Informed consent must account for how these data practices are communicated, understood, and agreed to by users. It is also important that users understand how their data is stored *after* the interaction. Unique consent challenges arise due to robots' embodied presence, multifunctional roles, and ability to gather data in ways that are not always visible or expected by users [7], [1], [2]. This is perhaps the most addressed dimension in current regulations (e.g., [3], [4], [5]) and is thoroughly addressed in regulations for research (e.g., [31]). However, robots complicate these regulations as their various roles may not be transparent, and data collection is often hidden and embedded in seemingly social or mundane interactions [7], [2]. When users are unaware of this dual role, it raises serious ethical concerns related to autonomy, trust, and privacy. Underestimating a robot's capabilities, such as not knowing that it is recording audio or video, can lead to significant breaches of privacy and feelings of betrayal when users realize the extent of the data collected. Additionally, people may stop questioning these practices altogether over time and experience consent fatigue [15], [24].

B. CREASE Principles

While the *components* in our RoboConsent framework provide aspects relevant for informed consent, the *principles* provide characteristics of informed consent and actionable steps towards it. Together, they ensure that each component of consent is, not only covered in content, but also delivered in a manner that is ethical, user-centered, and effective. These principles serve as a practical guide for shaping the process of informed consent throughout an HRI interaction.

Drawing inspiration from feminist consent models [15], [16], [23], we present the following six principles that we believe the consent process should adhere to: Continuous, Reversible, Enthusiastic, Adaptable, Specific, and Explainable. We call these principles CREASE, reflecting both the content of the principle and a nod to its purpose of *ironing out the CREASEs* in informed consent practices. While models like FRIES [23] were developed to guide sexual consent between individuals, CREASE builds on these ideas

to target issues specifically related to HRI like explainability, consent fatigue, and users' unmet expectations of robots. CREASE is therefore designed with the unique dynamics of HRI in mind while staying rooted in the feminist foundations that models like FRIES offer. The principles aim to move beyond traditional, one-time, and one-dimensional consent practices that are often insufficient in HRI. Each principle is outlined below:

1) **Continuous consent:** Consent should be treated as an ongoing process that continues throughout, and even after, the interaction and should not be viewed as a one-time event. In HRI, this means checking in with users regularly and remaining attentive throughout the interaction. For example, as users gain more experience with a robot, their expectations may shift (*Component 1 – expectations and experiences*); what may have felt acceptable at the beginning might no longer feel appropriate after certain behaviors or changes in the interaction. This is also critical when a robot's function or purpose changes during the interaction (*Component 3 – purpose and function*), such as, if a robot initially acts as a companion but suddenly begins asking personal health questions. This shift may alter what users expect regarding data collection and privacy (*Component 5 – data and privacy*). If consent is not continuous, the result may be discomfort or a breakdown in trust. A continuous consent process helps identify these moments, allowing for clarification before any negative experiences can occur.

2) **Reversible consent:** Users must be able to withdraw consent at any point without facing negative consequences. This includes the ability to stop the interaction, retract previously shared information, or change their level of engagement with the robot. Reversibility goes hand in hand with continuous consent, because continuous check-ins allow opportunities to withdraw consent. This is important in, for example, situations when the robot's role involves authority over the user (*Component 3 – purpose and function*), which can create pressure to comply. If a robot is used in a workplace context (*Component 4 – context of interaction*), users may feel obligated to participate in fear of risking their jobs. Similarly, if the robot begins collecting unexpected data (*Component 5 – data and privacy*) or demonstrating more advanced capabilities than the user initially expected (*Component 1 – expectations and experiences*), the user must be able to reverse their consent.

3) **Enthusiastic consent:** Consent should be given voluntarily and with active willingness to the specific interaction. This principle focuses on the users' agency in the interaction, where users should feel confident and comfortable interacting with the robot, without pressure. As with previous examples, this principle is important in interactions such as when the robot appears to hold authority over the user (*Component 3 – purpose and function*), especially in contexts that reinforce power imbalances (*Component 4 – context of interaction*). A robot's capabilities (*Component 2 – capabilities and limitations*), particularly those that have deceptive or persuasive design to encourage social behaviors can subtly pressure users into consenting, even when they do not fully understand

what they are agreeing to (*Component 1 – expectations and experiences*). Therefore, ensuring that individuals genuinely want to engage with the robot is just as important as ensuring they can withdraw consent.

4) **Adaptable consent:** Consent should be flexible to each user's individual needs and the specifics of the interaction. This means adapting the consent process based on the user's expectations (*Component 1 – expectations and experiences*), the robot's capabilities (*Component 2 – capabilities and limitations*) and purpose (*Component 3 – purpose and function*), the context in which the interaction takes place (*Component 4 – context of interaction*), and how data may be collected and used (*Component 5 – data and privacy*). For example, if a user is familiar with voice assistants like Siri, the consent process should explain how the robot differs, what the robot's intended role is, whether it has multiple functions, and how those functions relate to its capabilities and data practices. Consent must also account for how the situation may shift or how the user's experience might change over time.

5) **Specific consent:** Consent should clearly cover particular aspects of the interaction, rather than being merely a blanket agreement. This principle focuses on the tailoring of the consent. Each relevant aspect of the interaction should be addressed individually to ensure that users understand exactly what they are consenting to. Vague or generalized consent undermines the process and can lead to misunderstandings or misplaced trust. This principle stresses the need for the informed consent to be specific across all components. For example, users should be informed about the robot's specific capabilities and limitations (*Component 2 – capabilities and limitations*), what function the robot has (*Component 3 – purpose and function*), and what kind of data will be collected (*Component 5 – data and privacy*). Being specific in these ways helps users make informed decisions without having to infer or assume anything incorrectly regarding the interaction.

6) **Explainable consent:** The robot and the interaction should be understandable to users in real time. This principle focuses on how comprehensible the consent is and includes transparency about what the robot is doing and why, while using language that is understandable for the users. Explainability is critical for building trust and helping users make informed decisions throughout the interaction. Users must be able to understand *how* the robot's capabilities work (*Component 2 – capabilities and limitations*), not just what it can do. The robot's purpose (*Component 3 – purpose and function*) must also be clearly communicated to avoid misinterpretation or misplaced trust. Users should know how data is collected and why (*Component 5 – data and privacy*). Explainable consent processes ensure that users are not just told what will happen but actually comprehend what is happening and why, in real time.

IV. DISCUSSION

In this paper, we have outlined the informed consent challenges specific to HRI, where traditional models of informed consent fall short. We argue that meaningful informed consent must address the unique and challenging dynamics of HRI, including power imbalances, deception, data practices, and misaligned expectations. We contribute to the ongoing discussion of informed consent with a framework that reimagines consent practices to address the unique challenges HRI faces. We call this framework RoboConsent, emphasizing the need for robust, robot-specific approaches to informed consent in HRI. This framework consists of five components, grounded in the specific issues raised in HRI and highlighted by previous authors [2], [1], [33], [7], [12], [17], and six principles, based on feminist informed consent models [23], [16]. Because RoboConsent accounts for users' individual experiences and expectations, it also supports users who may not fit normative assumptions in HRI design (e.g., people with disabilities or older adults). Ultimately, this framework aims to provide a structured way to approach informed consent that takes user needs and the complexity of HRI into account.

Future work should test and refine this framework through real-world implementation and evaluation. Including users in the further development of this framework is especially important, in line with participatory and co-design approaches [33]. A central practical question for future exploration concerns *who* or *what* should obtain informed consent. Since many robots are intended to act autonomously, we also envision that this framework could be embedded within the robot itself, at least to some degree, so that robots can perform their function even when a responsible human is not present. We believe this is important as the envisioned future of robots is that they will act autonomously in shared environments with humans without constant human intervention [13]. At the same time, it is important to recognize that having the robot fully administer informed consent may create certain expectations that affect the user experience, and may not always be appropriate, particularly as robots' social and human-like behaviors can mislead users about their actual capabilities [26]. Issues with power imbalance may also arise in such situations [17]. This question is therefore deeply rooted in the contextual nature of HRI, and it ultimately requires careful navigation of how and when consent is understood and given. A possible solution could be, as Lintvedt [7] suggests, to have a two-step consent model where a professional obtains the consent followed by the robot doing the same. The involvement of both a person and a robot supports several principles of the RoboConsent framework, including *continuous consent* by revisiting it at different stages both by the robot and a person, and supports *adaptable* and *explainable consent* by offering different ways to give information. For example, the person could explain the robot's *capabilities* without evoking high expectations that could arise if the robot delivered the same information.

In addition, future work should also consider how existing

informed consent practices, particularly those related to data and privacy, can be applied or adapted together with this framework. While current regulations such as GDPR [3] and PIPEDA [4] address data and privacy, they often treat consent as a standalone event and do not account for data collection by a robot that may not always be obvious for users [2]. RoboConsent builds on these protections by also addressing how users' expectations, the robot's capabilities and functions, and the surrounding context shape the consent process. These components are not isolated, but interdependent, and must be considered together to ensure that consent in HRI is not only legally compliant but also ethically meaningful. An important next step will be to examine how existing regulatory frameworks can be implemented within RoboConsent to ensure that consent practices are both ethically robust and legally sound.

Finally, further research is needed to explore how to balance the flexibility of the RoboConsent framework with practical implementation. While we argue that informed consent in HRI should be ongoing and situational, there are questions about how to do this without overwhelming users (e.g., inducing consent fatigue [24]) or complicating the interaction. We acknowledge that the presented framework is quite extensive, though we believe these components are closely related and can be addressed together in practice. In addition, while comprehensive, these components reflect issues already identified as critical by previous researchers, including the need to address expectations, contextual consent, and power dynamics [12], [17], [1]. Future work should explore how these principles and components can be adapted and simplified for use in real-life HRI scenarios, and to what extent users find them meaningful and usable. This may also include exploring how the framework could guide more ethically grounded consent processes in experimental contexts where researchers may need to temporarily withhold full disclosure to preserve study integrity. In such cases, RoboConsent may help design more participant-centered approaches in these constraints.

With this work, we contribute to the growing discourse on informed consent in HRI by offering a framework that addresses the field's ethical challenges for obtaining meaningful informed consent. The RoboConsent framework provides a foundation for ethically grounded, user-centered design that recognizes the realities of robots interacting with people in various settings. By moving beyond checkboxes and static forms, this framework repositions consent as an active, evolving process that centers users, challenges power imbalances, and aligns with the envisioned future of robots.

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